

sbintuitions/**sarashina-embedding-v1-1b**

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Sentence Similarity



sentence-transformers



Safetensors



Transformers



11 datasets



Japanese



English

llama

feature-extraction



text-generation-inference



arxiv:2409.07737



Train

Deploy

Use this model



Model card



Files



xet



Community

Downloads last month

1,759



Safetensors

Model size

1.22B params

Tensor type

F32

Files info

Inference Providers

Sentence Similarity

This model isn't deployed by any Inference Provider.



Ask for provider support

Datasets used to train sbintuitions/sarashina-embedding-v1-1b

Space using sbintuitions/sarashina-embedding-v1-1b 1

Sarashina-Embedding-v1-1B

日本語のREADME/Japanese README

"Sarashina-Embedding-v1-1B" is a Japanese text embedding model, based on the 1.2B-parameter Japanese LLM "Sarashina2.1-1B". We trained this model with multi-stage contrastive learning. We achieved the state-of-the-art average score across 16 datasets in JMTEB (Japanese Massive Text Embedding Benchmark).

This model maps sentences & paragraphs to a 1792-dimensional dense vector space and can be used for semantic textual similarity, semantic search, paraphrase mining, text classification, clustering, and other applications.

🔗 Model Details

🔗 Model Description

- **Model Type:** Sentence Transformer
- **Base model:** [Sarashina2.1-1B](#)
- **Maximum Sequence Length:** 8,192 tokens
- **Output Dimensionality:** 1,792 dimensions
- **Similarity Function:** Cosine Similarity
- **Language:** Japanese
- **License:** [Sarashina Model NonCommercial License Agreement](#)

🔗 Full Model Architecture

```
SentenceTransformer(  
  (0): Transformer({'max_seq_length': 8192, 'do_lower_case': False}) with Transfo  
  (1): Pooling({'word_embedding_dimension': 1792, 'pooling_mode_cls_token': False,  
  )
```

🔗 Usage

First install the Sentence Transformers library:

```
pip install -U sentence-transformers
```

Then you can load this model and run inference.

```

from sentence_transformers import SentenceTransformer

# Download from the 🤗 Hub
model = SentenceTransformer("sbintuitions/sarashina-embedding-v1-1b")
# Run inference
sentences = [
    '更級日記は、平安時代中期に菅原孝標女によって書かれた回想録です。',
    'Sarashinaは、SB Intuitionsが開発した日本語大規模言語モデルです。これまでに7B, 13B, 70B',
    'サラシナエンベディングは日本語言語モデルをベースにした日本語埋め込みモデルです。'
]
embeddings = model.encode(sentences)
print(embeddings.shape)
# [3, 1792]

# Get the similarity scores between the embeddings
similarities = model.similarity(embeddings, embeddings)
print(similarities.shape)
# [3, 3]

```

Note

- You do not need to add prefixes such as "Query: " and "Document: " to the beginning of the input sentence.
- This model is licensed under the [Sarashina Model NonCommercial License Agreement](#), which has restrictions on commercial use. If you are interested in utilizing this model for your business, please feel free to contact us through our [contact page](#).

🔗 Training

"Sarashina-Embedding-v1-1B" is created through the following two-stage learning process:

🔗 Stage 1: Weakly-supervised Learning

To achieve generic text embedding performance across a wide range of domains, we performed contrastive training on weakly-supervised data consisting of our own web-crawled data and open data.

[🔗 Datasets](#)

dataset	counts
Auto Wiki QA/NLI	50,521,135
web-crawled data (ours)	47,370,649
MQA	12,941,472
llm-japanese-dataset	9,074,340
Wikipedia	5,555,212
Quiz dataset (ours)	988,478
Natural Questions	132,796
JSQuAD	62,859
SNOW(T15+T23)	62,758
JaQuAD	31,746
MKQA	3,318
total	126,744,763

[🔗 Step2: Supervised Fine-tuning](#)

To enable the model to learn a more accurate query-document similarity, we performed supervised fine-tuning using the following datasets.

[🔗 Datasets](#)

dataset	counts
JSNLI	141,388

dataset	counts
<u>NU-MNLI</u>	67,987
<u>Mr. TyDi</u> (only Japanese subset)	3,697
<u>Natural Questions</u> (sampled)	20,000
total	233,072

🔗 Evaluation Results with JMTEB

Model	Max Tokens	Avg.	Retrieval	STS	Classification	Reranking	Clustering	PairCla
<u>OpenAI/text-embedding-3-large</u> ^[^oai]	8191	74.05	74.48	82.52	77.58	93.58	53.32	62.35
<u>cl-nagoya/ruri-large</u>	512	73.31	73.02	83.13	77.43	92.99	51.82	62.29
<u>pkshatech/GLuCoSE-base-ja-v2</u>	512	72.23	73.36	82.96	74.21	93.01	48.65	62.37
<u>pkshatech/RoSEtta-base-ja</u>	1024	72.04	73.21	81.39	72.41	92.69	53.23	61.74
<u>intfloat/multilingual-e5-large</u>	512	70.90	70.98	79.70	72.89	92.96	51.24	62.15
<u>Sarashina-Embedding-v1-1B</u> (This model)	8192	75.50	77.61	82.71	78.37	93.74	53.86	62.00

🔗 License

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[^oai]: Benchmarked on April 23, 2024.